

MERNA KARAM ABDALLAH

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PROFESSIONAL SUMMARY

Fourth-year Computer Science and Artificial Intelligence student at Benha National University (GPA 3.66/4.00) with hands-on experience across the full Machine Learning pipeline: data preprocessing, feature engineering, model training, hyperparameter tuning, and evaluation. Applied Statistics and Linear Algebra foundations to build Machine Learning, Deep Learning, NLP, Computer Vision, and Robotics projects using Python, C++, SQL, TensorFlow, Scikit-learn, Pandas, NumPy, and OpenCV. Seeking an internship opportunity in AI, Machine Learning, or Data Science.

EDUCATION

Benha National University — Cairo, Egypt | 2023 – 2027

Bachelor of Computer Science and Artificial Intelligence

Fourth Year | GPA: 3.66/4.00

WORK HISTORY

Machine Learning Trainee | DEPI | Jul 2026 – Present

- Training in applied Machine Learning as part of the Digital Egypt Pioneers Initiative program.

AI & Data Analysis Trainee | TechTrek | 2026

- Worked with data preprocessing, feature engineering, model training, and evaluation using Python and Scikit-learn.
- Received hands-on training in Machine Learning and Artificial Intelligence.

AI Trainee | Huawei HCIA Training | 2025

- Worked with data preprocessing, data analysis, and Machine Learning concepts using Python.
- Developed practical skills in Artificial Intelligence and Data Analysis.
- Completed training covering Machine Learning, Deep Learning, and Computer Vision.

PROJECTS

Job Posting Fraud Detection | NLP

- Designed and trained an NLP-based classification system to detect fraudulent job postings, applying text preprocessing, tokenization, and feature extraction.
- Built and evaluated Logistic Regression, Random Forest, and XGBoost classifiers on 3,576 test samples; Random Forest achieved 98% accuracy and a 78% F1-score for fraudulent job postings, while Logistic Regression achieved 81% recall for the fraudulent class.

E-Commerce Customer Purchase Prediction | Machine Learning

- Built an end-to-end Machine Learning pipeline to predict customer purchase behavior from e-commerce data, covering data preprocessing, feature engineering, model training, and evaluation.
- Applied statistical analysis and Python (Pandas, NumPy) to clean, explore, and prepare features for modeling.
- Trained and evaluated multiple classification models using Scikit-learn, achieving 85% accuracy, 51% precision, 74.9% recall, and an F1-score of 59.7%.

Sign Language Digit Classification | Neural Networks

- Developed and trained a Convolutional Neural Network (CNN) in TensorFlow for sign language digit classification, achieving 98% training accuracy, 88% validation accuracy, and 93% test accuracy.
- Iterated on model architecture and hyperparameters to improve classification accuracy on the sign language digit dataset.

Self-Driving Robot Simulation | Robotics

- Developed a self-driving robot simulation applying robotics and autonomous-navigation concepts to sensor-based decision-making.
- Processed simulated environmental and sensor data to support real-time movement decisions.

Image Processing Project

- Applied image processing and transformation techniques using Python and OpenCV for image analysis.

E-Commerce Website | Front-End Development

- Developed an e-commerce website using HTML, CSS, and JavaScript with responsive, user-focused layouts.

TECHNICAL SKILLS

Programming: Python, C++, SQL

Mathematics & Statistics: Linear Algebra, Statistics, Probability, Descriptive & Inferential Statistics

Machine Learning: Scikit-learn, Data Preprocessing, Feature Engineering, Model Training, Model Evaluation, Hyperparameter Tuning

Deep Learning: Neural Networks, CNN, TensorFlow

NLP: Natural Language Processing, Text Preprocessing, Text Classification

Data Science: Pandas, NumPy, Data Analysis

Computer Vision: OpenCV, Image Processing, Image Classification

Robotics: Robotics, Autonomous Systems, Self-Driving Robot Simulation

Web Development: HTML, CSS, JavaScript

Tools: Git, GitHub, Jupyter Notebook, Google Colab

Office Tools: Microsoft Excel, Microsoft Word, Microsoft PowerPoint

TRAINING & CERTIFICATIONS

- HCIA — Huawei Certified ICT Associate | Huawei | 2025
- Front-End Development Route | 2024
- Convolutional Neural Networks (CNN) | 2026
- TechTrek Training | 2026

SOFT SKILLS

Team Leadership | Communication | Teamwork & Collaboration | Problem Solving | Time Management | Adaptability | Critical Thinking

LANGUAGES

Arabic: Native | English: Very Good